Twinfield Web services manual

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important message

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publish information

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Table of contents

important message 2
contact information 2
publish information 2
Table of contents 3

Preface 5

Audience and Prerequisites 5
Using the Twinfield Web services 6
Schematic program flow 6
General Xml recommendations 7
Error handling 7
Field definitions 8
Attributes in the response Xml 8

Security 9

Session Login 10

Resources 10
Classes 10
Enumerations 10
Methods 11
Logon 11
SmsSendCode 12
SmsLogon 12
ChangePassword 13
Example (C#) 15

Single Sign On Login 18

Process Xml messages 18

Resources 18
ProcesXmlString 18
ProcessXmlDocument 19
ProcessXmlCompressed 20
Supported Xml formats 21

Twinfield Finders 24

Resources 24
Methods 24
Search 24
Finder types 25
Search fields 26
Finder Options 27
FinderData Class 29
Message Class 29
MessageType enumeration 30

Twinfield Hierarchy master 32

Resources 32

the no. 1 in online accounting

Classes 32
Enumerations 32
Methods 32

Matching 36

Resources 36
UndoMatch 36

Example SOAP Envelopes 38

SOAP Envelope: Login 38
SOAP Envelope: ProcessXml 39
SOAP Envelope: Abandon 40
SOAP Envelope: Finder 41

the no. 1 in online accounting

Preface

Audience and Prerequisites

This document is intended for developers who are designing or developing Twinfield Web service enabled applications.

The Twinfield User Interface (GUI) corresponds directly with the Twinfield Xml layer. When you need to build an integration with Twinfield we strongly advice to 'simulate' the interface (e.g. dimension settings and posting journals) manually in Twinfield. The (error) messages generated by the Twinfield client are the same as the error messages returned in the Web services response Xml messages.

Readers should have a working knowledge of the Twinfield concepts and HTTP, Xml, SOAP or Web services communication standard. General information on Xml, SOAP or web services can be found on http://www.w3schools.com

Note: Twinfield has an extensive education program to enable developers to design, create, integrate, and extend applications connected with Twinfield. Visit http://www.twinfield.com/en/education/ to learn more about the available education programs.

Standard compliance

The Twinfield Web services is implemented to comply with the following specifications:

specification	more information
SOAP 1.1	http://www.w3.org/TR/2000/NOTE-SOAP-20000508/
SOAP 1.2	http://www.w3.org/TR/soap12-part1/
WSDL 1.1	http://www.w3.org/TR/wsdl
WS-I Basic Profile 1.1	http://www.ws-i.org/Profiles/BasicProfile-1.1-2004-08-24.html

Development platforms

The Twinfield Web services are working with current SOAP development environments. In this document, we provide examples C# (.NET). To see a complete list of sample code, go to the F.A.Q. section in Twinfield, keyword C#, .Net, VB.Net, ASP.Net, PHP, Webservices, SOAP or Xml.

Developer account

A developer account can be requested by sending an email to: partners@twinfield.com. Please describe in this email your business case.

Support

In case of support issues your questions can be sent to Twinfield Support by using the Twinfield Online Support function in Twinfield. After raising the question to Twinfield Support you will receive an email with a ticket number. By using the Global Twinfield online support function all raised questions and answers can be tracked online.

Related documents

Documentation of all available Twinfield Xml messages are described in the corresponding Xml manuals. These manuals can be downloaded from the F.A.Q. in Twinfield, with the keywords: Webservices, SOAP or Xml.

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Using the Twinfield Web services

The Twinfield web services are split on two types of web servers:

- Login server
- Cluster servers

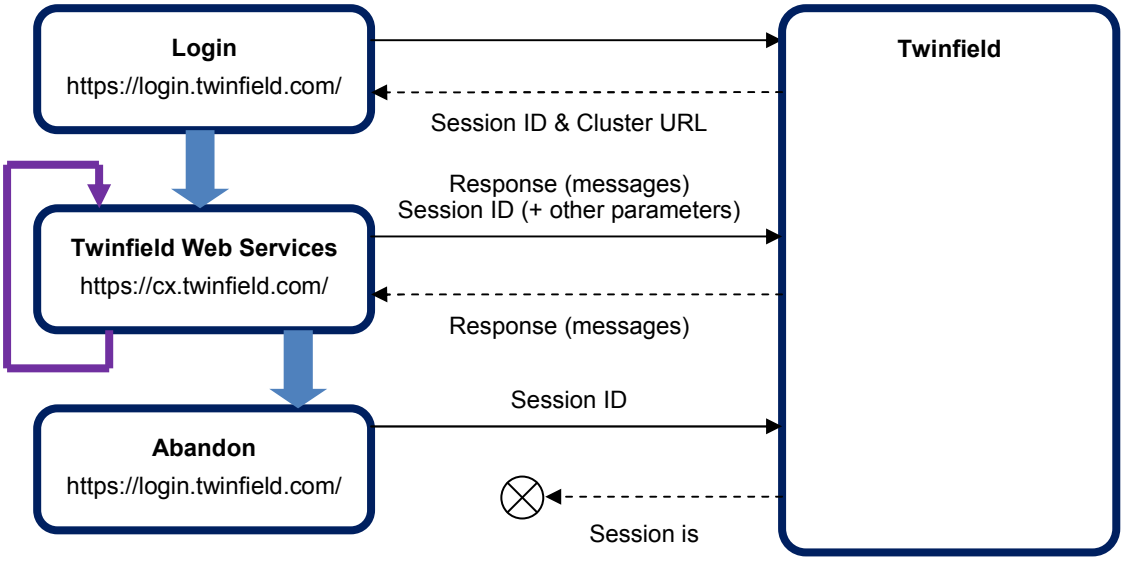
To use the Twinfield web services you first need to logon to the login server. To access an organization via the Twinfield Web services, you must first authenticate the session by sending a request to the login server at the following URL: https://login.twinfield.com/

In case of an authenticated user login, the login server will return a session ID and the URL to the correct organization's cluster server. An error will be raised when a web service request is made to a different cluster as returned by the login server. A session ID is valid for 20 minutes, every web service request will re-enable the session for 20 minutes.

NOTE: Per user only one (web services) session is allowed.

Schematic program flow

1. Start a new session.
   Web Services endpoint: https://login.twinfield.com/webservices/session.asmx
2. When the user credentials are correct the Session Web Service returns the session ID in the SOAP header and one of the return parameters of the Session Web Services is the cluster URL
   for example: https://cx.twinfield.com.
3. Based on the cluster URL the Twinfield Web Services functions can only be executed on the provided cluster server. For example the endpoint for ProcessXml is:
   https://cx.twinfield.com/webservices/processxml.asmx
4. Do other Web Services request on the same cluster.
5. Abandon your session
   Web Services endpoint: https://login.twinfield.com/webservices/session.asmx or the cluster



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General Xml recommendations

The Twinfield Web services are based on Xml, which requires all documents to be well formed. If an Xml document is not well formed it will raise an error event.

Xml Encoding

Xml documents may contain foreign characters, like Norwegian æ ø å , or French ê è é.
To let your Xml parser understand these characters, you should use the proper encoding which corresponds with your Xml documents, e.g. “ISO-8859-1”

```
<?xml version="1.0" encoding="ISO-8859-1"?>
```

Illegal Xml characters have to be replaced by entity references.

If you place a character like “<” inside an Xml element, it will generate an error because the parser interprets it as the start of a new element. You cannot write something like this:
There are 5 predefined entity references in Xml:

Xml character	Character	Description
<	<	Less than
>	>	Greater than
&	&	Ampersand
'	‘	Apostrophe
"	“	Quotation mark

Note: Only the characters “<” and “&” are strictly illegal in Xml. Apostrophes, quotation marks and greater than signs are legal, but it is a good habit to replace them.

Error handling

Twinfield errors are reported in the form of a ‘result’ attribute on the tags. If a posting is successfully executed, the root-tag of the Xml-Document shows the attribute ‘result’ with value ‘1’. If a problem occurs during the posting, the problem tag will show the attribute ‘result’ with value ‘0’. In addition two extra attributes are shown, namely ‘msg’ with a description of the problem and ‘msgtype’. The tags which are hierarchically ‘above’ this tag will also have the attribute ‘result’ with value ‘0’ (this also applies to the root-tag).

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Field definitions

Certain “tags” are of a fixed format, such as date and amount fields.

Type	Format	Description
BOOLEAN	true/false	True or Not true
DATE	yyyymmdd	ISO date format
MONEY	0.00	Full stop used as decimal separator.
PERIOD	yyyy/mm	Year/period
STRING(XX)	Free text field with maximum xx characters	
FIXED VALUE	The element value is fixed to the given value.	
CODE	Restricted text field with maximum of 16 characters. The allowed characters are: • “A” to “Z” (uppercase), • “0” to 9, • “_” (underscore), • “-“ (hyphen), • “.” (dot), • “,” (comma) and • “+” (plus).	Twinfield code field. This field will be checked if the corresponding master is available in Twinfield.
INTEGER	Numeric value, zero or greater	
ENUMERATION	Restricted value set	The values allowed are described in inverted commas with | used as the separator

After processing the Xml Twinfield will populate in the result Xml, extra attributes on the CODE fields. For example the dimension tags, ‘name’ and ‘shortname’ with the name and search name of the dimension.

Attributes in the response Xml

The following attributes can be added in the response Xml.

Type	Format	Description
result	Result of the posting 0 = not successful, 1= successful	ENUMERATION (1|0)
msg	Notification about why the posting failed	STRING
msgtype	Notification type; warning or error	ENUMERATION (“warning|error”)
name	Name of the Twinfield master	STRING
shortname	Short name of the Twinfield master	STRING
type	Type of dimension (only for dimensions)	STRING
inuse	Have financial transactions been posted to this dimension	BOOLEAN (“true|false”)
vatcode	Reserved; only for dimensions	
vatobligatory	Reserved; only for dimensions	

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Security

Client applications that access the Twinfield environment data are subject to the same security protections that are used in the Twinfield user interface.

User Authentication

Client applications must log in using valid Twinfield credentials for the Twinfield environment. The Twinfield server authenticates these credentials and, if valid, provides the client application with:

- a Session ID which must be set into the session header so that all subsequent calls to the Twinfield Web service are authenticated.
- a URL address (cluster) for future client application's Twinfield Web service requests.

Twinfield supports only the Secure Sockets Layer (SSL) protocol SSLv3 and the Transport Layer Security (TLS) protocol. Ciphers must have a key length of at least 128 bits.

Firewall settings

This chapter describes the firewall settings to enable the Twinfield Web services/SOAP.

IP-addresses:

- In range from: 194.0.194.150 to 194.0.194.159

Ports:

- 443 (SSL). Towards Twinfield 443 as DST. The web browser or SOAP client can connect via the SRC port >1024 (to 65.535 max.).

It is very important that content scanners, like IDP's, content inspection tools or other web tools, allow Xml and SOAP traffic and don't block these messages.

Note: We strongly recommend not to use intermediate servers e.g. Microsoft ISA server, which will decrypt the SSL traffic and encrypt the packages for the proxy server(s).

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Session Login

This chapter describes the functional description of the Twinfield Session web service.

Resources

resource	location
Web Service	/webservices/session.asmx
WSDL File	/webservices/session.asmx?wsdl

Classes

Header

The session header is a SOAP header, containing a session identifier. When the *Logon* method is executed successfully, it has created the session header itself and the session identifier has been assigned. In case of all other session method calls, the session header must be assigned before making the call.

```
public class Header : SoapHeader
{
  public string SessionID
}
```

Example

```
// Create instance of session web service.
TwinfieldSession.Session session = new TwinfieldSession.Session();

// Create and assign the header.
session.HeaderValue = new TwinfieldSession.Header();
session.HeaderValue.SessionID = sessionID;
```

Enumerations

LogonAction

The enumerated log-on action type is used as an out parameter in several session methods to determine the next required log-on action.

```
public enum LogonAction
{
  None = 0,
  SMSLogon = 1,
  ChangePassword = 2
}
```

log-on action	description
None	No action required.
SMSLogon	The next required action is SMS authentication.
ChangePassword	The next required action is change of password.

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Methods

Logon

Log on to a new session by providing the user code, password and organization code. If the cluster value is assigned, all other web service calls need to be performed on this URL.

```
public enum LogonResult : int
{
  Ok = 0,
  Blocked = 1,
  Untrusted = 2,
  Invalid = 3,
  Deleted = 4,
  Disabled = 5,
  OrganisationInactive = 6
}

public LogonResult Logon(
  string user,
  string password,
  string organisation,
  out LogonAction nextAction,
  out string cluster
)
```

Parameters

parameter	description
user	User code.
password	Password.
organisation	Organization code.
nextAction	Next required log-on action.
cluster	The required cluster URL for this log on.

Return value

Log-on result enumeration type.

log-on result	description
Ok	Log-on successful.
Blocked	Log-on is blocked, because of system maintenance.
Untrusted	Log-on is not trusted.
Invalid	Log-on is invalid.
Deleted	Log-on is deleted.
Disabled	Log-on is disabled.
OrganisationInactive	Organization is inactive.

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SmsSendCode

Send an SMS code to the session's user phone number. Only possible if the *LogonAction* is *SMSLogon*.

```
public int SmsSendCode()
```

Return value
If send was successful, the number of seconds in which the code times out is returned.

SmsLogon

Authenticate the session log-on by providing an SMS code.

```
public enum SMSLogonResult : int
{
  Ok = 0,
  Invalid = 1,
  TimeOut = 2,
  Disabled = 3
}
public SMSLogonResult SmsLogon(
  string smsCode,
  out LogonAction nextLogon
)
```

Parameters

parameter	description
smsCode	SMS code.
nextAction	Next required log-on action.

Return value
SMS log-on result enumeration type.

result	description
Ok	SMS log-on successful.
Invalid	SMS log-on is invalid.
TimeOut	SMS log-on has timed out.
Disabled	Log-on is disabled, because SMS log-on was invalid too many times.

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ChangePassword

Change the password of the current session's user by providing his current password (for security reasons) and the new password.

```
public enum ChangePasswordResult : int
{
  Ok = 0,
  Invalid = 1,
  NotDifferent = 2,
  NotSecure = 3,
  Disabled = 4
}

public ChangePasswordResult ChangePassword(
  string currentPassword,
  string newPassword,
  out LogonAction nextLogon
)
```

Parameters

parameter	description
currentPassword	Current password.
newPassword	New password.
nextAction	Next required log-on action.

Return value

Change password result enumeration type.

change password result	description
Ok	Change password successful.
Invalid	Current password is invalid.
NotDifferent	The current and new password are not different.
NotSecure	The new password is not secure.
Disabled	Log-on is disabled, because current password was invalid too many times.

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SelectCompany

Select the company for the current session by providing it's code.

```
public enum SelectCompanyResult : int
{
  Ok = 0,
  Invalid = 1
}

public SelectCompanyResult SelectCompany(
  string company
)
```

Parameters

parameter	description
company	Company code.

Return value

Select company result enumeration type.

result	description
Ok	Select company successful.
Invalid	Selected company is invalid or the user has insufficient rights to open it.

GetRole

Get the role of the currently logged in user.

```
public string GetRole()
```

KeepAlive

```
public void KeepAlive()
```

Keep the session alive, to prevent session time out. A session time out will occur after 20 minutes.

Abandon

Close the session.

```
public void Abandon()
```


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Example (C#)

```
// Create the session.
TwinfieldSession.Session session = new TwinfieldSession.Session();

// Request the user, password and organisation.
Console.WriteLine("User name:");
string user = Console.ReadLine();
Console.WriteLine("Password:");
string password = Console.ReadLine();
Console.WriteLine("Organisation:");
string organisation = Console.ReadLine();

// Log on to the session.
TwinfieldSession.LogonAction nextAction;
string cluster;
TwinfieldSession.LogonResult logonResult =
  session.Logon(user, password, organisation, out nextAction, out cluster);

// Check the cluster URL.
if (cluster != string.Empty)
{ session.Url = cluster + "/webservices/session.asmx"; }

// Act on the log-on result.
switch (logonResult)
{
  case TwinfieldSession.LogonResult.Ok:
    string actionError = string.Empty;
    while ((actionError == string.Empty) && (nextAction !=
TwinfieldSession.LogonAction.None))
    {
      switch (nextAction)
      {
        case TwinfieldSession.LogonAction.SMSLogon:
          // Send the code to the session user.
          int timeout = session.SmsSendCode();
          if (timeout > 0)
          {
            // Request the sms code.
            Console.WriteLine("An SMS has been sent, please fill in the code within {0}
seconds:", timeout);
            string code = Console.ReadLine();
            // Logon with the received sms code.
            TwinfieldSession.SMSLogonResult smsLogonResult =
              session.SmsLogon(code, out nextAction);
            switch (smsLogonResult)
            {
              case TwinfieldSession.SMSLogonResult.Invalid:
                actionError = "SMS log-on invalid.";
                break;
              case TwinfieldSession.SMSLogonResult.TimeOut:
                actionError = "SMS code timed out.";
                break;
              case TwinfieldSession.SMSLogonResult.Disabled:
                actionError = "Log-on is disabled.";
                break;
            }
          }
          else
          { actionError = "SMS failed to send"; }
```


the no. 1 in online accounting

```
            break;
          case TwinfieldSession.LogonAction.ChangePassword:
            // Request the passwords.
            Console.WriteLine("Current password:");
            string currentPassword = Console.ReadLine();
            Console.WriteLine("New password:");
            string newPassword = Console.ReadLine();
            // Change the password.
            TwinfieldSession.ChangePasswordResult changePasswordResult =
              session.ChangePassword(currentPassword, newPassword, out nextAction);
            switch (changePasswordResult)
            {
              case TwinfieldSession.ChangePasswordResult.Invalid:
                actionError = "Current password is invalid.";
                break;
              case TwinfieldSession.ChangePasswordResult.NotDifferent:
                actionError = "Passwords are not different.";
                break;
              case TwinfieldSession.ChangePasswordResult.NotSecure:
                actionError = "New password is not secure.";
                break;
              case TwinfieldSession.ChangePasswordResult.Disabled:
                actionError = "Log-on is disabled.";
                break;
            }
            break;
        }
      }
      if (actionError != string.Empty)
      { Console.WriteLine(actionError); }
      else
      { Console.WriteLine("Log-on successful."); }
      break;
    case TwinfieldSession.LogonResult.Blocked:
      Console.WriteLine("Log-on is blocked.");
      break;
    case TwinfieldSession.LogonResult.Untrusted:
      Console.WriteLine("Log-on is untrusted.");
      break;
    case TwinfieldSession.LogonResult.Invalid:
      Console.WriteLine("Log-on is invalid.");
      break;
    case TwinfieldSession.LogonResult.Deleted:
      Console.WriteLine("Log-on is deleted.");
      break;
    case TwinfieldSession.LogonResult.Disabled:
      Console.WriteLine("Log-on is disabled.");
      break;
    case TwinfieldSession.LogonResult.OrganisationInactive:
      Console.WriteLine("Organisation is inactive.");
      break;
    default:
      Console.WriteLine("Unknown log-on result.");
      break;
  }
```


the no. 1 in online accounting

Error Handling

If an error occurs in a web service method call, a SOAP exception is raised. The SOAP exception contains a detail Xml document.

```
<detail>
  <message>Error message</message>
  <code>100</code>
  <source>KeepAlive</source>
</detail>
```

The detail information contains the error message. This message is language dependent, therefor the detail information also contains a code value. This makes it possible to take predefined actions on specific error codes. Finally the source name is included, which contains the name of the called method.
error codes

The following error code are defined:

code	description
100	Unexpected exception.
101	Session header is missing.
102	Access is denied.
103	The log-on credentials are not valid anymore.
104	The log-on has been deleted.
105	The log-on has been disabled.
106	The organisation is no longer active.
107	SMS failed to send.
108	Access to this server is not allowed because the cluster is invalid.
109	You need access to at least one company to log on.
110	Login is not allowed on this server.

Error handling Example (C#)

```
try
{
  // Create instance of session web service.
  TwinfieldSession.Session session = new TwinfieldSession.Session();

  // Create and assign the header.
  session.HeaderValue = new TwinfieldSession.Header();
  session.HeaderValue.SessionID = sessionID;
}
catch (SoapException soapException)
{
  // Get the exception details
  string message = soapException.Detail.SelectSingleNode("message").InnerText;
  string code = soapException.Detail.SelectSingleNode("code").InnerText;
  string source = soapException.Detail.SelectSingleNode("source").InnerText;

  // Print the error.
  Debug.Print("SOAP Exception: {0}, error code: {1}, error source: {2}", message, code,
source);
}
```


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Single Sign On Login

The Twinfield Single Sign On Login Web services is described in the manual “Twinfield – Single Sign On”. This document can be downloaded from the F.A.Q. in Twinfield, keyword: SSO. In order to use Single Sign On the option must be enabled on your environment. This might require a signed addendum to your contract.

Process Xml messages

This chapter is a functional description of the Twinfield Process Xml web service, which can be used to exchange information with Twinfield.

The ProcessXml webservice contains the following functions:

- ProcessXmlDocument
- ProcessXmlString.
- ProcessXmlCompressed

Resources

resource	location
Web Service	/webservices/processxml.asmx
WSDL File	/webservices/processxml.asmx?wsdl

ProcesXmlString

```
public string ProcessXmlString(
        string xmlRequest
)
```

Methods

The sessionID must be sent as a SOAP header.

Parameters

parameter	description
xmlRequest	Xml request string.

Return value

Xml result string.

ProcessXmlString example (C#)

```
// Create instance of process xml web service.
TwinfieldProcessXml.ProcessXml processXml = new TwinfieldProcessXml.ProcessXml();

// Create and assign the header.
processXml.HeaderValue = new TwinfieldSession.Header();
processXml.HeaderValue.SessionID = sessionID;

// Create the request string.
string xmlRequest = string.Empty;
// ... fill the request string.
// Start processing the xml.
string xmlResult = processXml.ProcessXmlString(xmlRequest);
```


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ProcessXmlDocument

```
public XmlNode ProcessXmlDocument(
  XmlNode xmlRequest
)
```

Parameters

parameter	description
xmlRequest	Xml request node.

Return value
Xml result node.

Note: This method uses XmlNode instead of XmlDocument types. See http://support.microsoft.com/kb/330600 for additional details.

ProcessXmlDocument example (C#)

```
// Create instance of process xml web service.
TwinfieldProcessXml.ProcessXml processXml = new TwinfieldProcessXml.ProcessXml();

// Create and assign the header.
processXml.HeaderValue = new TwinfieldProcessXml.Header();
processXml.HeaderValue.SessionID = sessionID;

// Create the request document.
XmlDocument xmlRequest = new XmlDocument();

// ... fill the request document.

// Start processing the xml.
XmlDocument xmlDoc = new XmlDocument();
xmlDoc.LoadXml(processXml.ProcessXmlDocument(xmlRequest).OuterText);
```


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ProcessXmlCompressed

Processes a G-zipped UTF-8 Xml string and returns a G-zipped UTF-8 result Xml format. The G-zipped file is in the **base64Binary** type format.

```
public byte[] ProcessXmlCompressed(
  XmlNode xmlRequest
)
```

Parameters

parameter	description
xmlRequest	Xml request node.

Return value
Byte array containing a G-Zipped Xml String

ProcessXmlCompressed example (C#)

This example uses a class ZLib which is a wrapper around the zlib class from DotNetZip (http://www.codeplex.com/DotNetZip).

```
// Create instance of process xml web service.
TwinfieldProcessXml.ProcessXml processXml = new TwinfieldProcessXml.ProcessXml();

// Create and assign the header.
processXml.HeaderValue = new TwinfieldProcessXml.Header();
processXml.HeaderValue.SessionID = sessionID;

// Create the request document.
XmlDocument xmlRequest = new XmlDocument();

// ... fill the request document.

// Start processing the xml.
XmlDocument xmlDoc = new XmlDocument();

// Start processing the xml.
// Compress request
byte[] compressedRequest = Zlib.CompressString(xmlRequestString);
// Process request
byte[] compressedResponse = processXml.ProcessXmlCompressed(compressedRequest);
// Decompress response
string xmlResponseString = Zlib.DecompressString(compressedResponse);
```


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Supported Xml formats

The following types of Xml messages can be processed. The type is determined by the root element name.
If a container type is specified then a container type element may contain one or more process type elements to be processed. For example a <periods> root element may contain several <period> elements. The full Xml documentation and examples of the Xml types is described in several manuals. The column manual refers to the specific manual.

Type	Container type	Description	Manual
accountcode	accountcodes	Validate accounts	Validation
article	articles	Create of modify an article	Sales Invoice
assetmethod	assetmethods	Create or modify an asset method	Dimensions
cdmaction	cdmactions		Sales Invoice
columns		Returns transaction or hour data	Browse
currency	currencies	Create or modify currency data (incl. rates)	Currency
dimension	dimensions	Create or modify a dimension settings	Dimensions
dimensiongroup	dimensiongroups	Create or modify a dimension group	Dimensions
dimenstiontype	dimenstiontypes	Create or modify a dimension type	Dimensions
match		Perform matching	Matching
office	offices	Create or modify an office	Office
period	periods	Validate the period	Validation
projectrate	projectrates	Create or modify a project rate	Dimensions
remindruntype	remindruntypes		Sales Invoice
statement	statements	Post a bank statement in Twinfield	Transactions
salesinvoice	salesinvoices	Add sales invoice in Twinfield	Sales Invoice
transaction	transactions	Post a journal, purchase or sales invoice	Transactions
teq	teqs	Generation of projects and activities	Dimensions
user	users	Create or modify an user	User
vat	vats	Create or modify a VAT code	Vat
vatgroup	vatgroups	Create or modify VAT reporting group	Vat

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Special types

Type	Description
general	Container for different types. A general root element can contain any of the other elements listed. This way different process types can be combined to a single request.
list	Generate lists of several types of data. We recommend to use the Finder webservice to search for list items.
read	Reads several types of data

General element

A general root element can contain any of the other elements listed. This way different process types can be combined to a single request.

Example Xml

```
<general>
  <dimensiongroups/>
  <dimensions/>
  <vats/>
  <transactions/>
  <!-- ... -->
</general>
```

List element

The list type is specified in a type element.
For some types additional elements may or must be specified.

Type	Description
browsefields	Lists all available browse fields.
budgets	Lists budget information.
dimensions	Lists dimensions of a specific dimension type. Recommendation is to use the Finder webservice.
offices	Lists all offices which are available to the user. Recommendation is to use the Finder webservice.

Example Xml

```
<list>
  <type>offices</type>
</list>
```

Read element

The list type is specified in a type element. For some types additional elements may or must be specified. A full description and examples of the types can found in the “validation” manual (MAN-VALIDATE). This manual can be downloaded from the Twinfield F.A.Q.; keyword: Webservices, Xml or SOAP.

the no. 1 in online accounting

Type	Description
article	Returns the article definition of a particular article.
browse	Returns the browse definition of a particular.
dimensions	Returns the dimension definition of a particular dimension.
office	Returns the office settings
projectrate	Returns the project rate settings
salesinvoice	Returns the sales invoice details of an invoice.
transaction	Returns the transaction details of a particular transaction.
user	Returns the user settings
vat	Returns the VAT settings

Example: read article
This example reads the article settings of the article with: EXAMPLE.

```
<read>
  <type>article</type>
  <office>001</office>
  <code>EXAMPLE</code>
</read>
```

Example: read browse
This example reads the browse settings of the customer overview (100) in office 001.

```
<read>
  <type>browse</type>
  <office>001</office>
  <code>100</code>
</read>
```

Example: read dimensions
This example reads the dimension setting of customer (DEB) 1000 in office 001.

```
<read>
  <type>dimensions</type>
  <office>001</office>
  <dimtype>DEB</dimtype>
  <code>1000</code>
</read>
```

Example: read office
This example reads the office settings of office 001.

```
<read>
  <type>office</type>
  <code>001</code>
</read>
```

Example: read project rate
This example reads the project rate settings of project rate CONS in office 001.

```
<read>
  <type>projectrate</type>
  <office>001</office>
  <code>CONS</code>
</read>
```


the no. 1 in online accounting

Example: read sales invoice
This example reads sales invoice 20090001 from invoice type INVOICE in office 001.

```
<read>
  <office>001</office>
  <type>salesinvoice</type>
  <code>INVOICE</code>
  <number>20090001</number>
</read>
```

Example: read transaction
This example reads transaction 200900001 from daybook BNK in office 001.

```
<read>
  <type>transaction</type>
  <office>001</office>
  <code>BNK</code>
  <number>200900001</number>
</read>
```

Example: read user
This example reads the user settings of user EXAMPLE.

```
<read>
  <type>user</type>
  <code>EXAMPLE</code>
</read>
```

Example: read VAT
This example reads the VAT settings of VH in office 001.

```
<read>
  <type>vat</type>
  <office>001</office>
  <code>VH</code>
</read>
```

Twinfield Finders

This chapter describes the Twinfield finder Webservices to search for Twinfield masters.

Resources

resource	location
Web Service	/webservices/finder.asmx
WSDL File	/webservices/ finder.asmx?wsdl

Methods

The sessionID must be sent as a SOAP header.

Search

```
public Message[] Search(
    string type,
    string pattern,
    int field,
    int firstRow,
    int maxRows,
    string[][] options,
    out FinderData data)
```


the no. 1 in online accounting

The method will return an array containing validation messages. If there are no errors, then the data parameter will contain the actual search result.

Parameters

parameter	description
type	A string which specifies the finder type. The available finders are described in the chapter “Finder types”.
pattern	The search pattern. May contain wildcards * and ?.
field	Determines which field or fields will be searched. Allowed values depend on finder type. The available finder fields are described in the chapter “Finder types”.
firstRow	The index of the first row to return. One-based. Must be >= 1.
maxRows	The maximum number of rows to return. When set to 0 all rows will be returned. Must be >= 0.
options	An array containing additional options (finder type specific). May be left empty. The available finder options are described in the chapter “Finder options”.
data	Returns the search results.

Finder types

The following finders are available

type	Twinfield master / function	available fields
ART	Articles	code, name
ASM	Asset methods	code, name
BDS	Budgets	code, name
CDA	Credit management action codes	code, name
COL	Collect payment types	code, name
CTR	Countries	code, name
CUR	Currencies	code, name
DIM	Dimensions (financials)	code, name, bank account, address fields
DIM	Dimensions (modifiedsince option)	code, name, modified, revision
DIM	Dimensions (projects)	code, name, customer.
DMT	Dimension types	code, name
DVT	Vat types	code, name
FLT	Filter mappings	code, name
FMT	Payment files	code, name
GRP	Dimension groups	code, name
HIE	Hierarchies	code, name
HND	Hierarchy nodes	code, name
INV	Invoice types	code, name
IVT	List of available (not paid) invoices	Invoice number, relation code, relation name, amount, debit/credit

the no. 1 in online accounting

type	Twinfield master / function	available fields
MAT	Matching types	code, name
OFF	Offices	code, name
OFG	Office groups	code, name
OIC	Available offices for InterCompany transactions	code, name
PAY	Payment types	code, name
PTY	Pay and collect payment types	code, name
REP	Reports	code, name
REW	Word templates (invoices, dunning letters)	code, name
RMD	Dunning scenarios	code, name
ROL	User roles	code, name
SMP	Salary employees	code, name
SPM	Distribution by periods (Budgets/Extended Trial Balance allocation)	code, name
SPP	Salary employees per period	code, name
TEQ	Time & quantities transaction types	code, name
TRS	Transaction types (daybooks)	code, name
TRT	Time project rates	code, name
USR	Users	code, name
VAT	VAT codes	code, name
VGM	VAT Groups	code, name
XLT	Translations	code, name

Search fields

The search field determines which field of fields will be searched. The available fields depend on the finder type. Passing a value outside the specified values will cause an error.

finder type	description	field code
all	Searches on the code or name field	0
all	Searches only on the code field	1
all	Searched only on the name field	2
DIM	Searches in for a dimension bank account number	3
DIM	Searches in the dimensions address fields	4
DIM	Searched for the customer code (only available for projects/activities)	3
IVT	Searches for the dimension code or dimension name, outstanding invoice amount, invoice number	0

the no. 1 in online accounting

Finder Options

The Finder options Passing an unsupported name or value causes an error. An option name may be used once, specifying an option multiple times will cause an error.

type	option name	option value	description
all	office	«office code»	Sets the office code
ART	vat	“inclusive” “exclusive”	Return only articles configured with a value including or excluding the VAT.
CTR	ctrtype	“vatnl”	Return only countries that can be used for Intra-Community Supplies in a Dutch VAT declaration.
DIM (financials)	level	“1”, “2”, “3” “4”, “5”, “6”	Specifies the dimension level. This option is mandatory when section = "teq". When the section = “teq” only level “1” or “2” are available
	section	“financials”	Restricts to financial or time & expenses dimensions.
	dimtype	“BAS”, “PNL”, “CRD”, “DEB”, “KPL”, “PRJ”, “AST”, “ACT”	Specifies the dimension type.
	modifiedsince	yyyyMMddHHmmss or yyyyMMdd	Restricts to dimensions modified after, on or at the specified date (and time).
	group	«group code»	Specifies the dimension group (wildcards allowed).
	matchtype	"relation"	This option will show only the relations (dimension type: “CRD” and/or “DEB”).
DIM (time &billing)	accessrules	“0”, “1”	Disabling the access rules is only possible if the user has access to project dimension maintenance. This option is only available when the section is “teq”.
	accessuser	«user code»	Specifies the time user. This option is only available when the section is “teq”.
	prefix		Specifies that the search field should start with the given value. This option is only available when the section is “teq”.
	role	«userrole code»	Specifies the time user role. This option is only available when the section is “teq”.
	trscode	«transaction code»	Specifies the transaction code and will only return the projects or activities which are related to the specified transaction code.
DMT	dim	«dimension code»	Restricts to the type of the specified dimension.
	finlevel	“1”, “2”, “3” “4”, “5”, “6”	Specifies the Financial level.
GRP	prefix		Specifies that the search field should start with the given value.
	dimtype	“BAS”, “PNL”, “CRD”, “DEB”, “KPL”, “PRJ”, “AST”, “ACT”	Specifies the dimension type. The result data will only show the dimension groups of the specified dimension type.
HIE	officegroup	officegroup code	Sets the office group
HIE	accestype	"use", "bind", "edit"	Specifies the minimal required access type for the reporting structure

the no. 1 in online accounting

type	option name	option value	description
HND	accestype	"use", "bind", "edit"	Specifies the minimal required access type for the reporting structure
HND	hierarchy	reporting structure code	Specifies the reporting structure
INV	vat	"inclusive", "exclusive"	Return only invoice types configured for including or excluding VAT.
IVT	dim1 – dim6	«dimension code»	Specifies a dimension code (and financial level).
	value	«x.yy»	Specifies the (original invoice) value.
	openvalue	«x.yy»	Specifies the outstanding value.
OFF	accessrules	“0”, “1”	Disabling the access rules is only possible if the user has access to the office master.
	consolidate	“0”, “1”	Disabling the access rules is only possible if the user has access to the office master.
PAY	paytype	"pay", "collect"	Specifies which type of pay and collect types should be returned.
REW	paytype	"pay", "collect"	Specifies the purpose of the report / template.
SPP	payperiod	«pp»	Specifies the employee salary payment period
	period	«pp»	Specifies the employee salary period
	subperiod	«pp»	Specifies the employee salary sub-period
	year	«pp»	Specifies the salary year
TRS	hidden	“1”	Include “hidden” transaction codes in the result set
	ic	“1”	Shows only “InterCompany” transactions in the result set. This option cannot be used in combination with the 'hidden' option.
TRT	ratetype	“time”, “quantity”	Specifies the rate type
USR	accessrules	"0", "1"	Disabling the access rules is only possible if the user has access to user management.
	mutualoffices	"0", "1"	Shows all users who have access to a selection of offices that overlaps with the Web Service user permissions.
VAT	vattype	"purchase", "sales"	Specifies the vat type.

Passing an unsupported name or value causes an error. An option name may be used once, specifying an option multiple times will cause an error.

the no. 1 in online accounting

Access rules option

If accessrules = 1, the normal access rules to the items in the finder are applied. E.g. the office finder will only show the offices where the user has access to. If accessrules = 0, no access rules will be applied. This makes all items visible. The access rules option is enabled (1) by default and may only be disabled (0) if the user has access to the maintenance section of the items.

FinderData Class

The FinderData class is a container for the search results.

```
public class FinderData {

  public int TotalRows {
    get { }
    set { }
  }

  public string[] Columns {
    get { }
    set { }
  }

  public string[][] Items {
    get { }
    set { }
  }
}
```

Parameters

parameter	description
TotalRows	The total number of search results.
Columns	The column names
Items	The requested data items

Message Class

```
public class Message {
  public MessageType Type {
    get { }
    set { }
  }
  public string Text {
    get { }
    set { }
  }
  public int Code {
    get { }
    set { }
  }
  public string[] Parameters {
    get { }
    set { }
  }
}
```


the no. 1 in online accounting

Parameters

parameter	description
Type	The type of the message.
Text	The message text.
Code	The error code, see below for possible error codes.
Parameters	Optional parameters related to the error.

Error codes

code	cause
1	You don't have access to office «office code».
2	Option «option name» is not allowed
3	Option «option name» must be 0 or 1.
4	Option «option name» must be an integer value.
5	Option «option name» must be a decimal value.
6	Option «option name» must be one of «list of values».
7	Option «option name» must be between «lower value» and «upper value».
8	Parameter «parameter name» must be between «lower value» and «upper value».
9	Finder type «finder type» is not supported.
10	Parameter «parameter name» must be greater than or equal to «value».
11	Option 'level' is mandatory when section = 'teq'
12	Option 'ic' cannot be used in combination with option 'hidden'.

MessageType enumeration

```
public enumeration MessageType {
  Warning,
  Error
}
```


the no. 1 in online accounting

Finder example (C#)

```
// Create instance of finder webservice
TwinfieldFinder.Finder finder = new TwinfieldFinder.Finder();
// Set sessionID header
finder.HeaderValue = new TwinfieldFinder.Header();
finder.HeaderValue.SessionID = session.HeaderValue.SessionID;
// Build options array
string[][] options = new string[2][];
options[0] = new string[] { "dimtype", "DEB" };
options[1] = new string[] { "section", "financials" };
// Search dimensions of type "DEB" with bank account which starts with "1*",
// return the first 10 matches.
TwinfieldFinder.FinderData results;
TwinfieldFinder.MessageOfErrorCodes[] messages;
messages = finder.Search("DIM", "1*", 3, 1, 10, options, out results);
   if (messages.Length != 0)
   {
      // Print errors
      foreach (TwinfieldFinder.MessageOfErrorCodes message in messages)
      {
         Console.WriteLine(message.Text);
      }
   }
   else
   {
      // Print results
      if (results.Items != null)
      {
         for (int i = 0; i < results.Items.Length; ++i)
         {
            Console.WriteLine("{0}: Customer {1} ({2})", i,
               results.Items[i][0], results.Items[i][1]);
         }
      }
      else
      {
         Console.WriteLine("Nothing found.");
      }
   }
Console.WriteLine("Press any key to continue...");
Console.ReadKey(); }
```


the no. 1 in online accounting

Twinfield Hierarchy master

This chapter describes the Twinfield hierarchy Webservice to maintain the hierarchy master.

Resources

resource	location
Web Service	/webservices/hierarchies.asmx
WSDL File	/webservices/hierarchies.asmx?wsdl

Classes

Classes	Description
Hierarchy	
HierarchyNode	
HierarchyAccount	

Enumerations

```
public enumeration HierarchyAccountBalanceType{
   Credit = -1,
   Balance = 0,
   Debit = 1
}
```

Methods

The sessionID must be sent as a SOAP header.

Classes	Description
Load	This method load the specified hierarchy code.
Save	This method saves the hierarchy.
Bind	This method can bind node to an hierarchy.
Unbind	This method can bind node to an hierarchy.
Hide	This method hides the specified hierarchy code.
Delete	This method deletes the specified hierarchy code.

the no. 1 in online accounting

Load

```
public Message[] Load(
   string hierarchyCode,
   out Hierarchy hierarchy
)
```

Parameters

parameter	description
hierarchyCode	User code.
Hierarchy	The out parameter to assign the loaded hierarchy to.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

Save

```
public Message[] Save(
   ref Hierarchy hierarchy
)
```

Parameters

parameter	description
hierarchyCode	User code.
Hierarchy	The out parameter to assign the loaded hierarchy to.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

the no. 1 in online accounting

Bind

```
public Message[] Bind(
  int node,
  string office,
  HierarchyAccount[] accounts
)
```

Parameters

parameter	description
node	The identification of the node to bind the accounts to.
office	The office code to add the bindings for.
accounts	An array of hierarchy accounts to bind to the node.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

Unbind

```
public Message[] Bind(
  int node,
  string office,
  HierarchyAccount[] accounts
)
```

Parameters

parameter	description
node	The identification of the node to bind the accounts to.
office	The office code to add the bindings for.
accounts	An array of hierarchy accounts to bind to the node.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

the no. 1 in online accounting

Hide

```
public Message[] Hide(
   string hierarchyCode
)
```

Parameters

parameter	description
hierarchyCode	The hierarchy code.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

Delete

```
public Message[] Delete(
   string hierarchyCode
)
```

Parameters

parameter	description
hierarchyCode	The hierarchy code.

Return value
An array containing validation messages.

log-on result	description
Message	An array containing validation messages.

the no. 1 in online accounting

Matching

This chapter is a functional description of the Twinfield Process Xml web service, which can be used to exchange information with Twinfield.

The Matching webservice contains the following functions:

- UndoMatch

Resources

resource	location
Web Service	/webservices/matching.asmx
WSDL File	/webservices/matching.asmx?wsdl

UndoMatch

```
public string UndoMatch(
        string office,
        string dimensionType,
        string dimension,
        int matchNumber
)
```

Methods

The sessionID must be sent as a SOAP header.

Parameters

parameter	description
office	The office (administration) code
dimensionType	The dimension type, the following values are allowed: - BAS for balance sheet accounts - PNL for profit & loss accounts - CRD for suppliers - DEB for customers - PRJ for projects The other dimension types are not allowed, because the “matching” cannot been set on the dimension
dimension	The dimension code
matchNumber	The match number

Return value

Xml result string.

the no. 1 in online accounting

UndoMatch example (C#)

```
// Create instance of process xml web service.
TwinfieldProcessXml.ProcessXml processXml = new TwinfieldProcessXml.ProcessXml();

// Create and assign the header.
processXml.HeaderValue = new TwinfieldSession.Header();
processXml.HeaderValue.SessionID = sessionID;

// Create the request string.
string xmlRequest = string.Empty;
// ... fill the request string.
// Start processing the xml.
string xmlResult = processXml.ProcessXmlString(xmlRequest);
```


the no. 1 in online accounting

Example SOAP Envelopes

This chapter is an example of the SOAP Envelopes to create a Twinfield session, process an Xml message and abandon the session.

SOAP Envelope: Login

The SOAP Envelope is posted to https://login.twinfield.com/webservices/session.asmx?wsdl

Request

```
<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Body>
    <Logon xmlns="http://www.twinfield.com/">
      <user>«usercode»</user>
      <password>«password»</password>
      <organisation>«organisation»</organisation>
    </Logon>
  </soap:Body>
</soap:Envelope>
```

Response

```
ResponseCode: 200 (OK)
Content-Length:549
Cache-Control:private, max-age=0
Content-Type:text/xml; charset=utf-8
Date:Tue, 24 Mar 2009 11:24:26 GMT
X-AspNet-Version:2.0.50727
X-Powered-By:ASP.NET

<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Header>
    <Header xmlns="http://www.twinfield.com/">
      <SessionID>61587139-7938-4dca-84f7-625524d7a8fa</SessionID>
    </Header>
  </soap:Header>
  <soap:Body>
    <LogonResponse xmlns="http://www.twinfield.com/">
      <LogonResult>Ok</LogonResult>
      <nextAction>None</nextAction>
      <cluster>https://c1.twinfield.com</cluster>
    </LogonResponse>
  </soap:Body>
</soap:Envelope>
```


the no. 1 in online accounting

SOAP Envelope: ProcessXml

The described SOAP Envelope uses the Twinfield ProcessXmlString function and this example requests returns a list of available offices. This SOAP envelope is posted to the cluster URL (returned by the session Envelope) and added with /webservices/processxml.asmx?wsdl. based on our example the request is posted to: https://c1.twinfield.com/webservices/processxml.asmx?wsdl

Note: The session ID must be included in the SOAP Header.

Request

```
<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Header>
    <Header MyAttribute="" xmlns="http://www.twinfield.com/">
      <SessionID>61587139-7938-4dca-84f7-625524d7a8fa</SessionID>
    </Header>
  </soap:Header>
  <soap:Body>
    <ProcessXmlString xmlns="http://www.twinfield.com/">
     <xmlRequest><![CDATA[<list><type>offices</type></list>]]></xmlRequest>
    </ProcessXmlString>
  </soap:Body>
</soap:Envelope>
```

Response

```
ResponseCode: 200 (OK)
Connection:close
Content-Length:598
Cache-Control:private, max-age=0
Content-Type:text/xml; charset=utf-8
Date:Tue, 24 Mar 2009 12:33:25 GMT
X-AspNet-Version:2.0.50727

<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Body>
    <ProcessXmlStringResponse xmlns="http://www.twinfield.com/">
      <ProcessXmlStringResult>&lt;offices result="1"&gt;&lt;office name="MORE&amp;amp;Zo
Services BV" shortname="Services BV"&gt;001&lt;/office&gt;&lt;office
name="More&amp;amp;Zo Holding"
shortname="Holding"&gt;010&lt;/office&gt;&lt;/offices&gt;</ProcessXmlStringResult>
    </ProcessXmlStringResponse>
  </soap:Body>
</soap:Envelope>
```


the no. 1 in online accounting

SOAP Envelope: Abandon

The SOAP Envelope to abandon the Twinfield session can be posted to https://login.twinfield.com/webservices/session.asmx?wsdl or to the ProcessXml web service on the cluster URL (in our example https://c1.twinfield.com/webservices/processxml.asmx?wsdl).

Note: The session ID must be included in the SOAP Header.

Request

```
<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Header>
    <Header MyAttribute="" xmlns="http://www.twinfield.com/">
      <SessionID>61587139-7938-4dca-84f7-625524d7a8fa</SessionID>
    </Header>
  </soap:Header>
  <soap:Body>
    <Abandon xmlns="http://www.twinfield.com/" />
  </soap:Body>
</soap:Envelope>
```

Response

```
ResponseCode: 200 (OK)
Content-Length:299
Cache-Control:private, max-age=0
Content-Type:text/xml; charset=utf-8
Date:Tue, 24 Mar 2009 11:27:30 GMT
X-AspNet-Version:2.0.50727
X-Powered-By:ASP.NET

<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XmlSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XmlSchema">
  <soap:Body>
    <AbandonResponse xmlns="http://www.twinfield.com/" />
  </soap:Body>
</soap:Envelope>
```


the no. 1 in online accounting

SOAP Envelope: Finder

The SOAP Envelope for the finder Web Services must be posted to the cluster URL (in our example https://c1.twinfield.com/webservices/finder.asmx?wsdl).

This example returns all Suppliers which are in office 0000.

Note: The session ID must be included in the SOAP Header.

Request

```
<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Header>
    <Header MyAttribute="" xmlns="http://www.twinfield.com/">
      <SessionID>61587139-7938-4dca-84f7-625524d7a8fa</SessionID>
    </Header>
  </soap:Header>
  <soap:Body>
    <Search xmlns="http://www.twinfield.com/">
      <type>DIM</type>
      <pattern>*</pattern>
      <field>0</field>
      <firstRow>1</firstRow>
      <maxRows>0</maxRows>
      <options>
        <ArrayOfString>
          <string>office</string>
          <string>0000</string>
        </ArrayOfString>
        <ArrayOfString>
          <string>dimtype</string>
          <string>CRD</string>
        </ArrayOfString>
      </options>
    </Search>
  </soap:Body>
</soap:Envelope>
```


the no. 1 in online accounting

Response

```
ResponseCode: 200 (OK)
Connection:close
Content-Length:908
Cache-Control:private, max-age=0
Content-Type:text/xml; charset=utf-8
Date:Thu, 10 Jun 2010 07:46:27 GMT
Server:Microsoft-IIS/7.5
X-AspNet-Version:2.0.50727

<?xml version="1.0" encoding="utf-16"?>
<soap:Envelope xmlns:soap="http://schemas.xmlsoap.org/soap/envelope/"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <soap:Body>
    <SearchResponse xmlns="http://www.twinfield.com/">
      <SearchResult />
      <data>
        <TotalRows>5</TotalRows>
        <Columns>
          <string>Code</string>
          <string>Naam</string>
        </Columns>
        <Items>
          <ArrayOfString>
            <string>2001</string>
            <string>Vodafone Ltd.</string>
          </ArrayOfString>
          <ArrayOfString>
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            <string>Heuvel</string>
          </ArrayOfString>
          <ArrayOfString>
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            <string>KvK</string>
          </ArrayOfString>
        </Items>
      </data>
    </SearchResponse>
  </soap:Body>
</soap:Envelope>
```


the no. 1 in online accounting